

Topic:-

Collect information about the material used in production of denture (a set of artificial teeth).

Introduction:-

Human teeth play an indispensable role in maintaining our health, nutrition, and social confidence. They allow us to bite and masticate solid food for smooth digestion, support our facial muscles, and enable us to pronounce words clearly. However, as people grow older, or due to severe tooth decay, gum diseases (pyorrhea), and accidental facial injuries, natural teeth are often lost.

Losing teeth makes it extremely painful and difficult for senior citizens to chew food, leading to malnutrition, sunken cheeks, and loss of self-confidence.

Fortunately, dental science solves this problem by manufacturing custom-made artificial sets of teeth called **Dentures** (बत्तीशी). A denture can be a complete set (replacing all 32 teeth) or a partial set (replacing a few missing teeth). Dentures are sophisticated medical prosthetics manufactured from advanced man-made materials. In Chapter 17 on Man-Made Materials, we studied the polymer chemistry and materials science behind modern dentures.

❖ Required Properties of Denture Materials:-

- **Biocompatibility:** Must be 100% non-toxic and non-irritating to sensitive mouth gums.
- **Mechanical Strength:** Must withstand biting and chewing pressures up to 200 kg/cm²!
- **Aesthetics:** Must mimic natural translucent white enamel and natural pink gum tissue.

Aim & Dental Research Plan:-

Aim: To collect, examine, and document information on the modern man-made materials, polymers, and ceramics used by dental surgeons and technicians to manufacture dentures.

❖ *Objectives of the Study:-*

- To identify the primary polymer used in pink denture bases (Acrylic PMMA resin).
- To compare porcelain ceramic teeth with composite cross-linked acrylic teeth.
- To study modern flexible nylon dentures and cobalt-chromium metal frameworks.

❖ *Information Sources Researched:-*

- Class 8 General Science textbook (Chapter 17: Man Made Materials).
- Interview and demonstration at a local dental laboratory and dental clinic.
- Indian Dental Association (IDA) Patient Education guides on Prosthodontics.



Manufacturing Process of Dentures:-

The dental technician demonstrated the multi-step precision fabrication of dentures:

- **1. Taking the Mouth Impression:** The dentist places soft alginate gel in a metal tray and presses it against the patient's gums to capture an exact replica of jaw contours.
- **2. Creating a Plaster Stone Cast:** Dental stone plaster is poured into the impression to create an exact stone working model of the patient's mouth.
- **3. Wax Rim Setup:** Technicians build pink beeswax rims and arrange artificial teeth individually, matching the patient's natural bite and smile line.
- **4. Acrylic Flasking & Polymerization:** The wax model is enclosed in a brass flask filled with plaster. Wax is boiled out, and the cavity is filled with pink Polymethyl Methacrylate (PMMA) acrylic resin dough. Heated under pressure, PMMA hardens into a rock-solid, smooth pink base.
- **5. Polishing & Delivery:** The denture is trimmed, pumice-polished to a high shine, and fitted into the patient's mouth.

Denture Materials Comparison Table:-

The table below compares the specific man-made materials used to fabricate modern dentures:

Denture Component	Material Used	Key Physical Property	Main Clinical Advantage
Pink Gum Base	Polymethyl Methacrylate (PMMA)	Lightweight acrylic polymer	Easily molded; matches natural pink gums
Artificial Teeth (Ceramic)	Dental Porcelain (Feldspar glass)	Extreme hardness & translucent shine	Natural white appearance; stain and scratch proof
Artificial Teeth (Resin)	Cross-linked Acrylic Composite	High impact fracture resistance	Does not click when talking; bonds chemically to base
Flexible Partial Base	Thermoplastic Nylon (Valplast)	Elastic, unbreakable flexibility	Comfortable fit without unsightly metal clasps
Cast Metal Frame	Cobalt-Chromium (Co-Cr) alloy	High tensile strength, ultra-thin	Extremely strong partial denture support

Scientific Analysis & Chemical Properties:-

Analyzing the chemistry of dental materials shows why PMMA remains the undisputed king:

- **Chemistry of PMMA Acrylic Resin:** Polymethyl Methacrylate is formed by polymerizing methyl methacrylate liquid monomers with powder polymers in the presence of a benzoyl peroxide catalyst. It is completely inert in saliva, resists mouth acids from citric fruits, and absorbs zero water.
- **Ceramic Porcelain vs Acrylic Teeth:**
 - *Porcelain Teeth:* Made from fused feldspathic glass heated to 1200°C. They look remarkably lifelike and never wear down from chewing. However, they are brittle and can chip if dropped on tiled floors.
 - *Acrylic Composite Teeth:* Softer and wear down slightly over years, but absorb chewing shock gently, protecting underlying jaw bones from stress.
- **CAD/CAM 3D Printed Dentures:** Modern dental labs now use digital intraoral laser scanners and 3D printing resins to design and print complete dentures within 2 hours!

Care, Hygiene & Social Impact:-

Proper maintenance is crucial for senior citizens using dentures daily:

- **1. Cleaning with Soft Brushes:** Dentures must be cleaned daily using mild soap and a soft-bristled brush to remove food debris. Never use abrasive toothpastes that scratch the acrylic shine.
- **2. Soaking Overnight:** Acrylic PMMA dentures must be removed at night and stored in a glass of clean water. If left to dry out in air, acrylic resin warps and loses its snug fit!
- **3. Restoring Dignity to Elders:** Beyond chewing food, dentures fill out facial cheeks, prevent wrinkles around the mouth, restore clear speech pronunciation, and give seniors back their proud, confident smiles.

❖ Caution Regarding Boiling Water:-

Never wash dentures in boiling water, as high temperatures soften thermoplastic PMMA and permanently warp its mouth shape!

What We Learned & Conclusion:-

- Dentures are custom-made dental prosthetics that replace missing teeth and gum tissues in senior citizens.
- Polymethyl Methacrylate (PMMA) acrylic resin forms the pink gum base due to its biocompatibility and molding ease.
- Dental porcelain ceramics and cross-linked resins create realistic, strong, and wear-resistant artificial teeth.
- Man-made materials and polymer chemistry directly restore nutrition, speech, health, and dignity to human lives.

Conclusion:-

Our study of denture materials provided an inspiring real-world demonstration of how synthetic man-made materials serve human healthcare. By engineering acrylic polymers, porcelain ceramics, and metal alloys, dental technicians recreate nature's dental anatomy with stunning precision. This project taught us that materials science is not just an industrial topic—it is a compassionate science that restores health, smiles, and happiness to our grandparents.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 17: Man Made Materials).
- Indian Dental Association (IDA) Guide to Removable Prosthodontics and Denture Care.
- Dental Materials Science: Principles and Applications, Balbharti Pune.